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Cambodia Renewable Energy & Policy Development

March 27, 2025

CONTENT:

1. **Energy Landscape in Cambodia**
   - *Progress of electrification*
   - *Energy demand & supply*
   - *How much energy is imported*
   - *Renewable energy progress*

2. **Cambodia's climate commitment and relevant energy policies**



Energy Landscape in Cambodia

❖ Energy Mix: Progress & Electrification

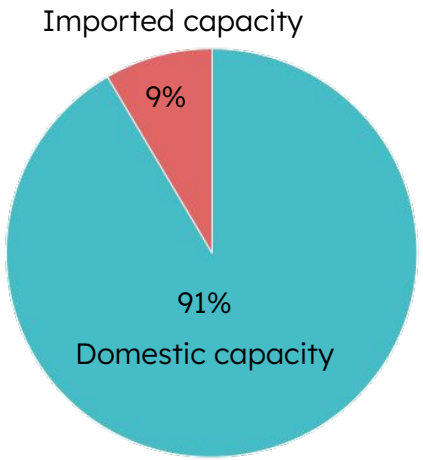
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Electrical Power (2003)

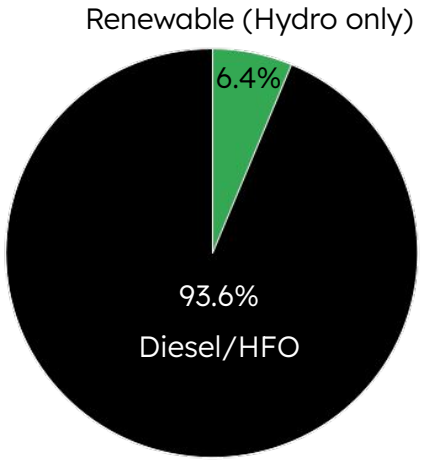
205 MW
Total Power Capacity

6.4%
RE Power Capacity

19.3%
Electrification Rate



Power Capacity



Domestic Power Capacity

Data: EAC, and MME

❖ Energy Mix: Progress & Electrification

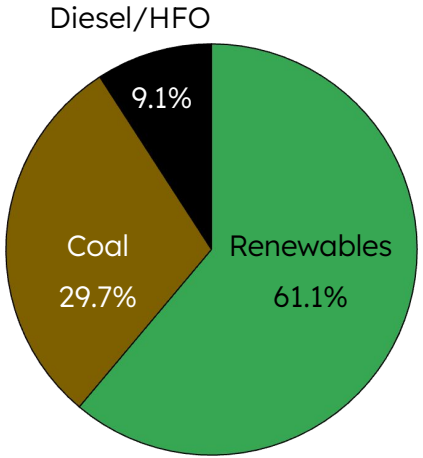
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Electrical Power (2024)

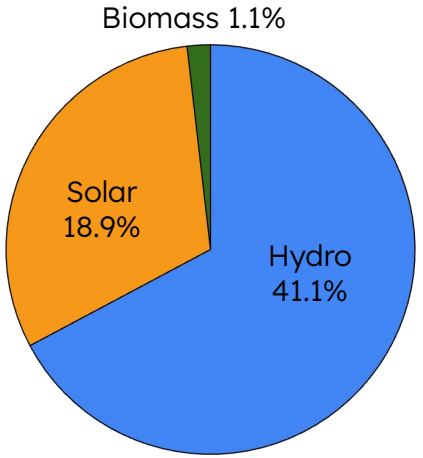
5,044 MW
Total Power Capacity

61%
RE Power Capacity

95%
Electrification Rate



Domestic Power Capacity

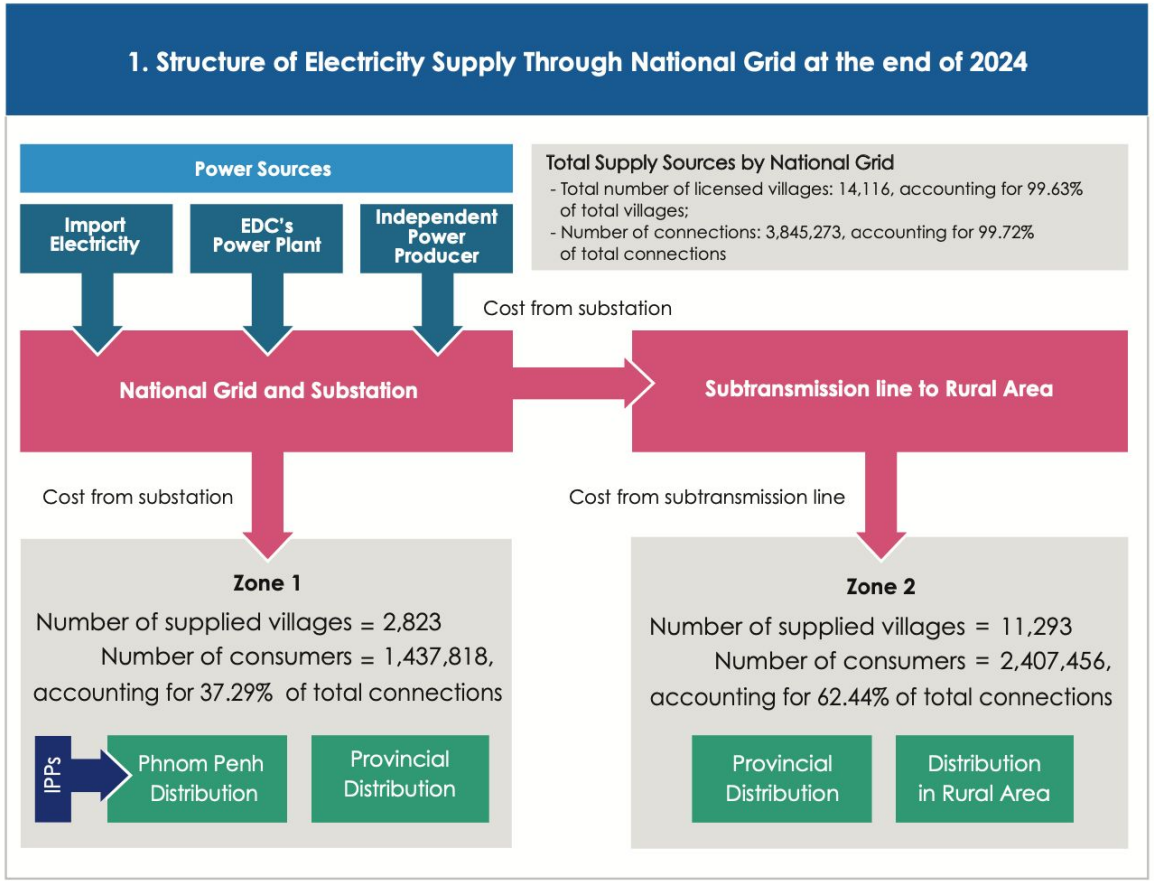


RE Power Capacity

Data: EAC, and MME

❖ Structure of Electricity Supply

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Electricity supply structure:

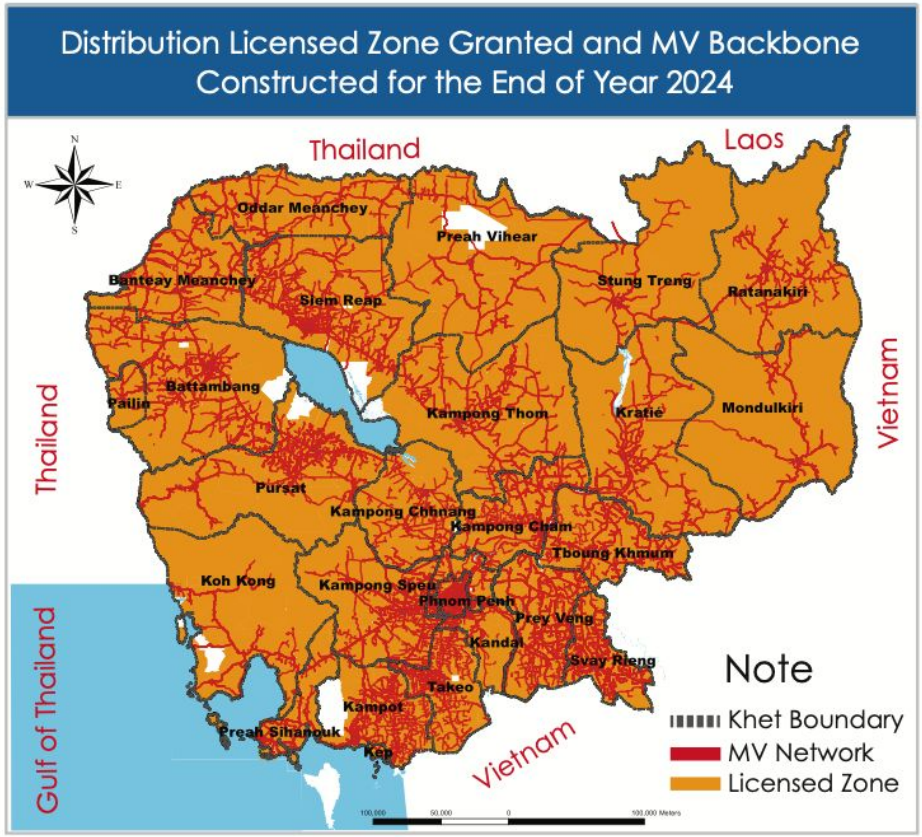
1. Supply through national grid
2. Small-scale imports
3. Mini-grid

Supply through national grid:

- Import electricity
- EDC's power plants
- Independent Power Producer (IPP)

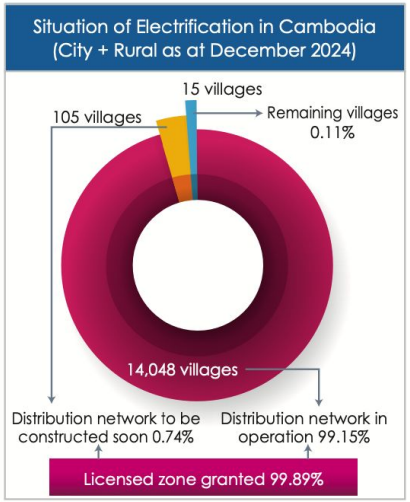
❖ Structure of Electricity Supply

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How much effort is needed for last-mile electrification?

What about quality, reliability, and affordability?

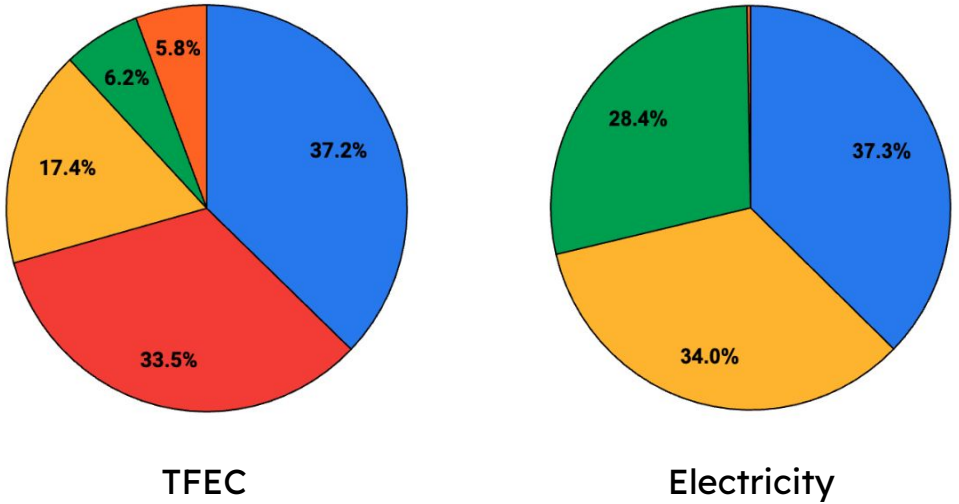


❖ Energy Demand by Sector

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Electricity and Total Final Energy Consumption (TFEC) by Sector in 2021

- Industrial
- Transport
- Residential
- Comm. & Public Services
- Others

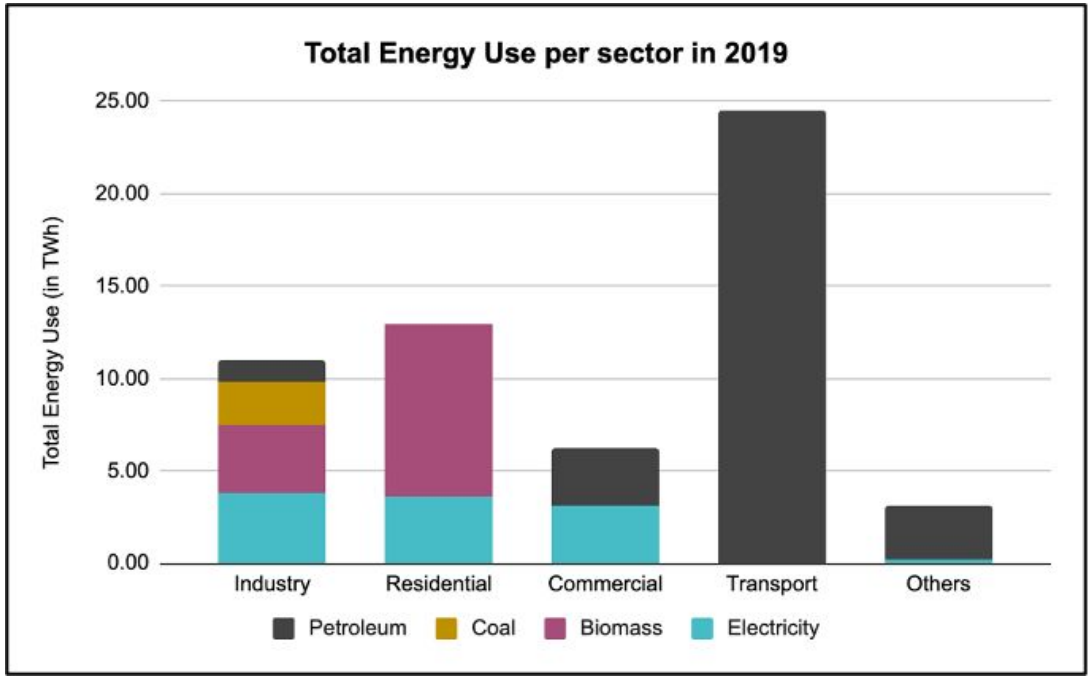


Statistics and Energy Balance 2020-2021, MME & UNDP

❖ Energy Demand by Sector

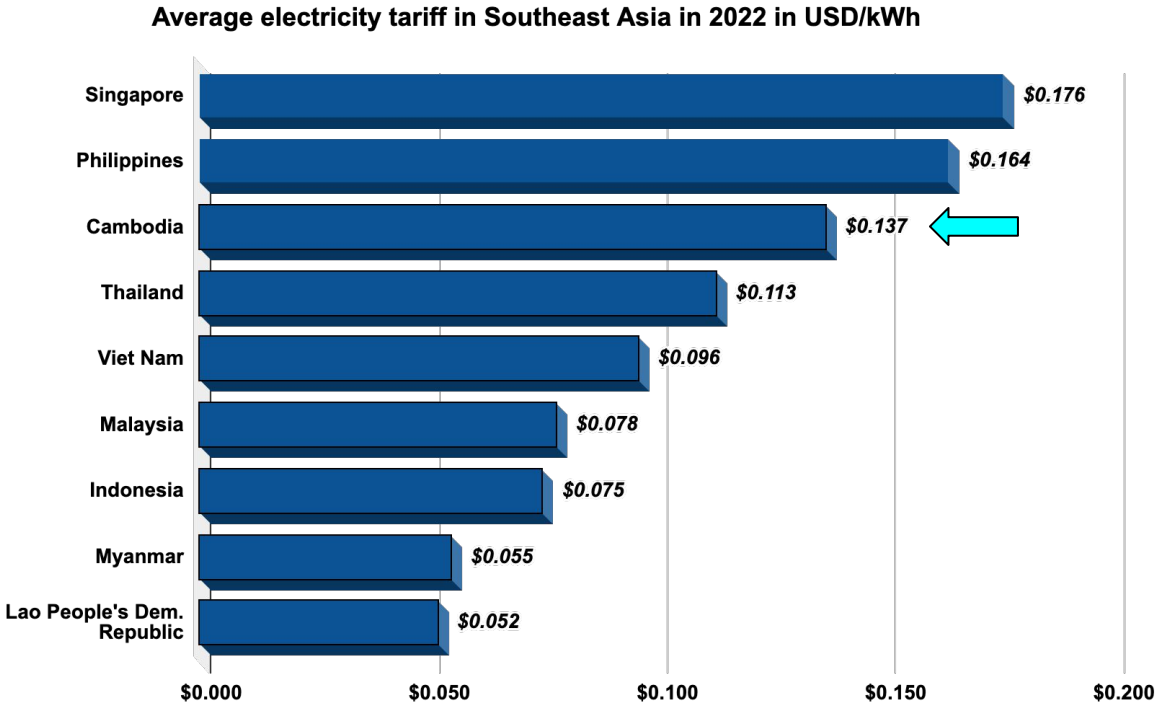
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Energy Use per Sector, 2019



❖ Electricity Price

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ELECTRICITY PRICE

Third highest in SEA!

USD 0.137/kWh

Source: ClimateScope.org



ELECTRICITY IMPORT (2024)

Imported 1,567 GWh of electricity from Laos, TH, VN.

At $0.07/kWh, it is estimated to be around **USD 110 million.**

Data from EAC + Author's calculation

Fossil Fuel Import

Diesel: $1,320 M
Petrol: $850 M



(2023)

Coal:
$443 M



(2022)

Natural gas & LPG:
$450 M



(2022)

Observatory of Economic Complexity (OEC)
https://www.phnompenhpost.com/business/oil-imports-dip-amidst-economical-green-shift

What is renewable energy?

Renewable energy is energy derived from natural sources that are replenished at a higher rate than they are consumed. Usually, renewable energy are considered clean, because they produce much less GHG than fossil fuels.

Renewable energy technologies: Solar, Wind, Hydro, Biomass, etc.



SOGE'S PV

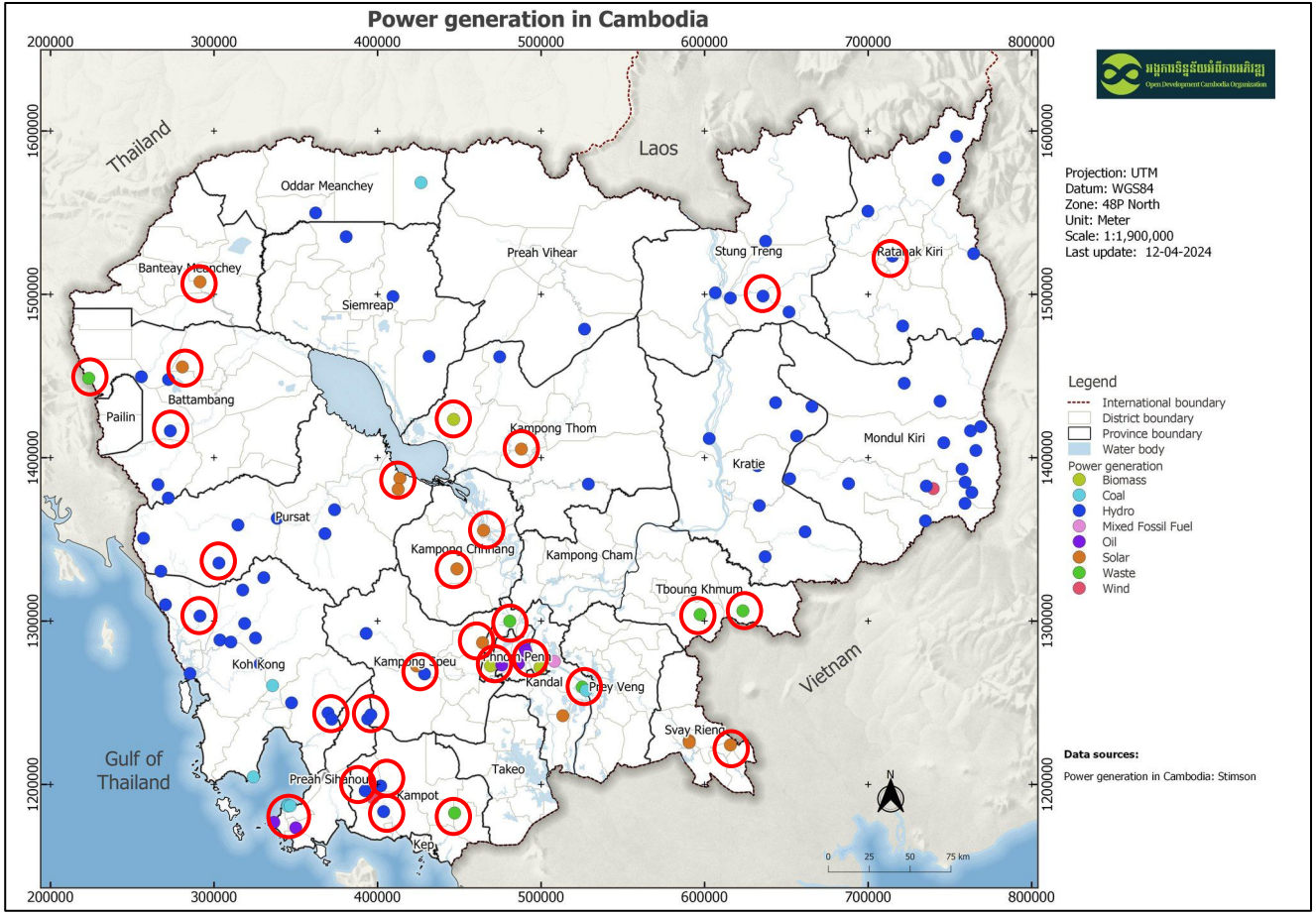






❖ Renewable Energy

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- **Hydropower**
- **Solar PV**
- **Biomass**

○ **Existing**

Source: Open Development Cambodia

❖ Renewable Energy

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Name - Province	Capacity (MW)	Year of Operation
Sunseap Solar Power - Svay Rieng	10	2017
Tai Seng Solar PV - Svay Rieng	5	2017
SchneiTech Solar Power Plant - K. Speu	80	2019
Soma Solar PV - K. Thom	5	2020
SchneiTech Sustainable - K. Chhnang	60	2020
SchneciTech Infinite - Pursat	30	2020
SchneiTech Sustainable - Pursat	60	2020
Solar Power Plant - B. Meanchey	39	2021
Ray Power - B. Meanchey	30	2021
Risen Energy - Battambang	60	2021
Green Sustainable Venture - Svay Rieng	20	2022
Prime Road Alternative - K. Chhnang	60	2022



Business July 20, 2018

ADB-backed solar park to be ready in November

Chea Vannak / Khmer Times

A solar farm in Bavet city, in Svay Rieng. KT/May Titthara



Risen Energy - Battambang (Google Map)

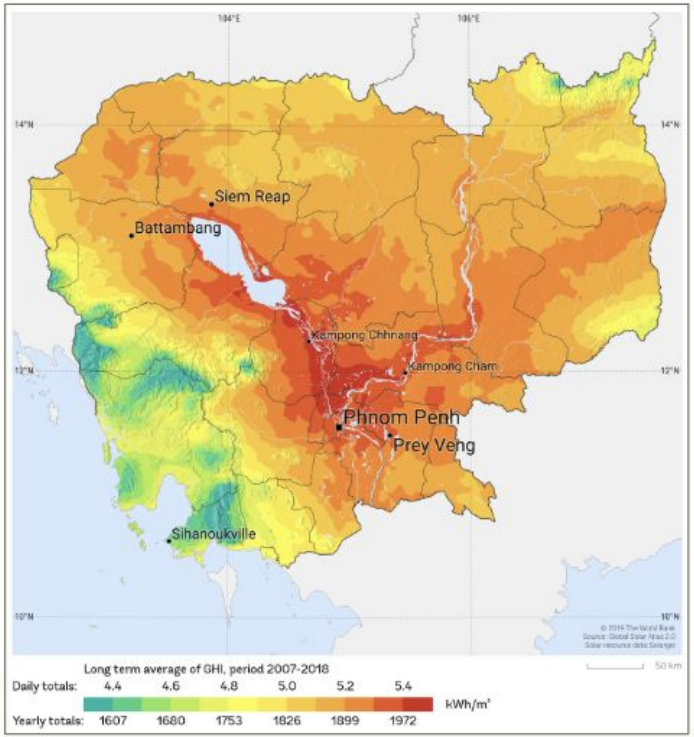


Prime Road Alternative
(Images from PRA & Khmer Times)

**Current projects more than on the list*

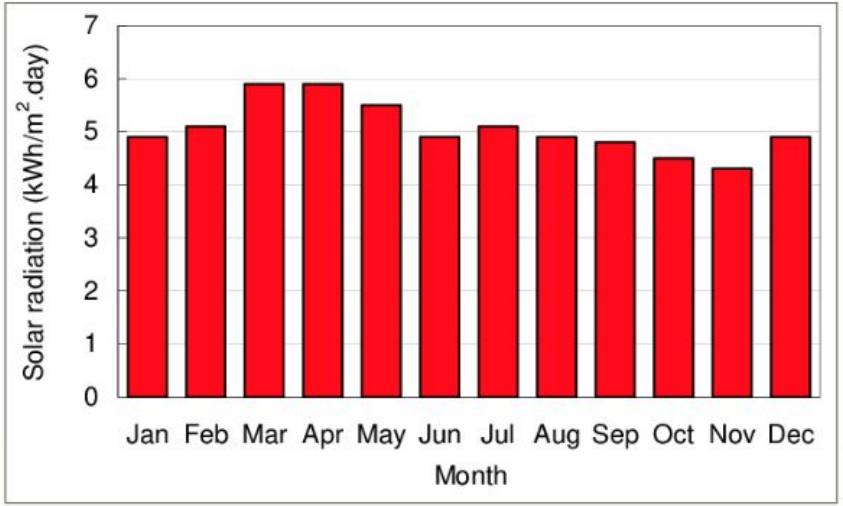
❖ Renewable Energy

(Energy)Lab



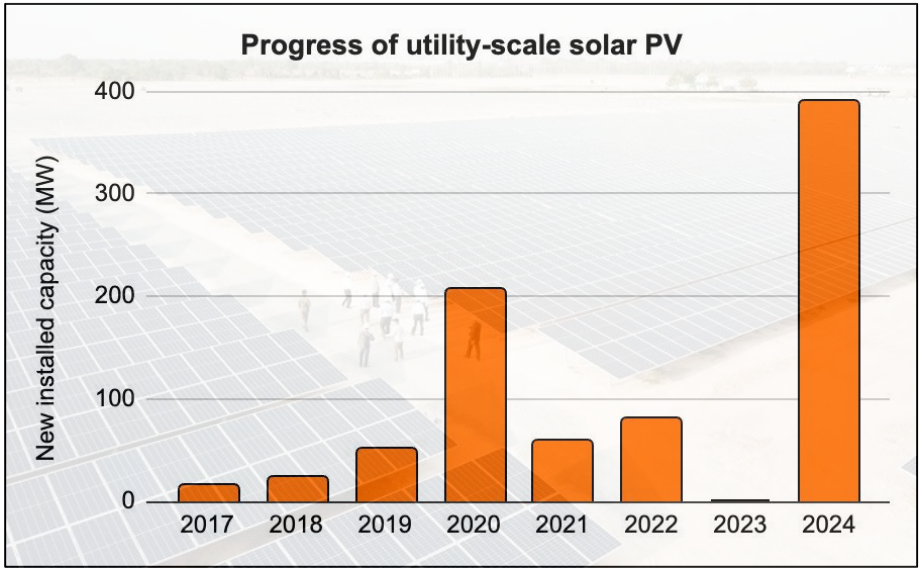
Source: Solargis

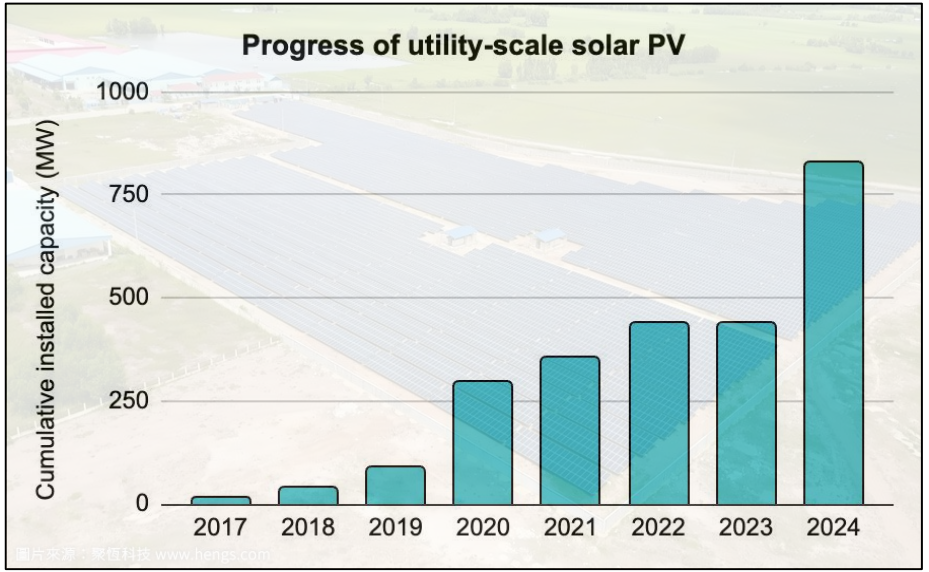
- ➔ Effective sunshine: 4-5 hour/day
- ➔ Average solar radiation: 5 kWh/m²/day
- ➔ Potential: 44 GW capacity



Ketjoy et al., 2007

Progress of Solar PV in Cambodia





Data from EAC Reports

Wind Power

- Early stage
- Limited potential
- Feasibility shows Kampot and Mondulkiri



Wind farm in Kampot
(Image source: The Phnom Penh Post)

1. Wind power project in Mondulkiri

- Capacity: 150 MW
- Status: Underway, scheduled to complete in 2026
- By: HK Oasis Power Energy Development

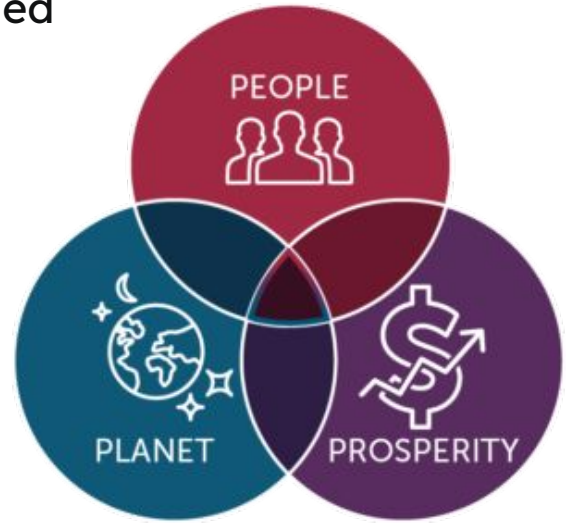
- Capacity: 150 MW
- Status: Agreement signed
- Scheduled to complete in 2027
- By: Leader Energy

2. Wind power project on Kampot

- Capacity: 80 MW
- Status: Completed? (With delayed PPA negotiation)
- By: The Blue Circle

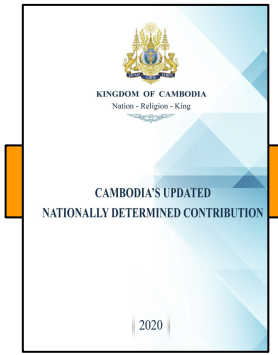
Why Renewable Energy? Why Solar?

- Economic and population growth > More energy needed
- Electricity access to rural and remote areas
- Climate change mitigation
- Support irrigation and agriculture
- Improve economic competitiveness
- Attract investment and improve economic
- Reduce reliance on fossil fuel > Energy security
- Factories meeting climate compliance



Cambodia’s climate commitment and relevant energy policies

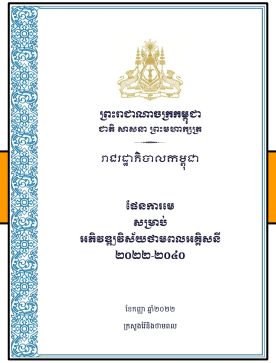
NATIONAL POLICIES TOWARD CARBON NEUTRALITY



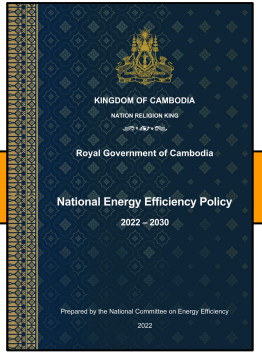
NDC 2020
CO2 Reduction
40% by 2030



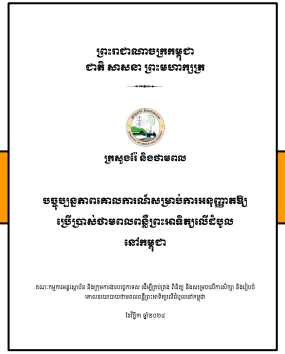
LTS4CN
Net Zero
by 2050



PDP
2022-2040
Max RE



NEEP
2022-2030
19% Reduction



RTS
Rooftop solar
In Cambodia

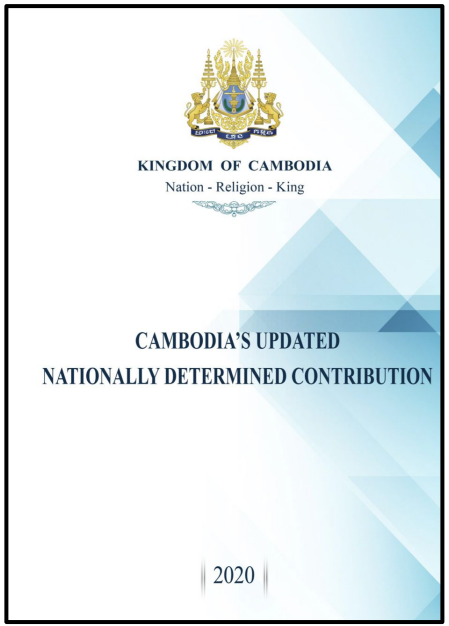


EVs 2024
Increase the
use of EVs

❖ Policy - NDCs 2.0

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NDCs submitted to United Nations Framework Convention on Climate Change (UNFCCC) as a national commitment to mitigate GHG and adapt to climate change impacts by 2030.



Mitigation Targets: Reduce GHG by **41.7%** compared to BAU by 2030:

- **Forestry and Other Land Use (FOLU):** ~50% of total reduction from the sector
- **Energy, agriculture, Industry, and Waste:** Remaining GHG reduction

Adaptation Strategies:

- **Infrastructure:** Enhance resilience to withstand climate impacts
- **Water resources:** Implementing sustainable water management practices
- **Agriculture:** Promoting climate-resilient agricultural techniques

The newer NDCs (NDCs 3.0) is being developed, and will be published in 2025.

❖ Policy - LTS4CN

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Long-Term Strategy for Carbon Neutrality (LTS4CN) 2021-2050 submitted to UNFCCC outlining the nation's commitment to achieving carbon neutrality by 2050, by reducing GHG across various sectors.

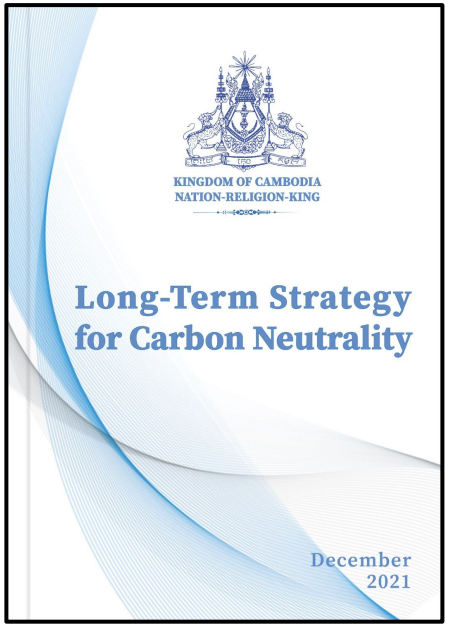


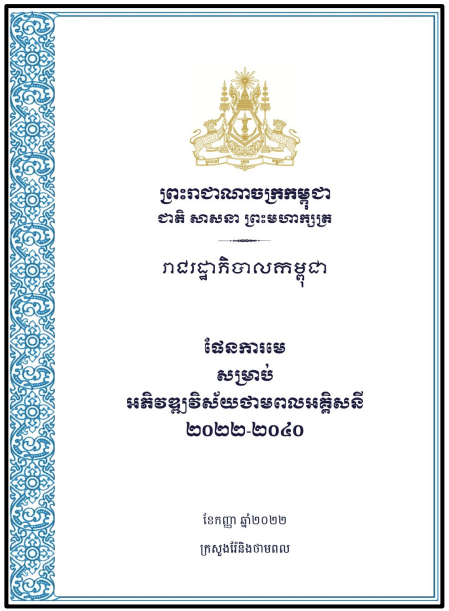
Table 2: GHG emissions projection by sector in 2050, BAU and LTS4CN scenarios

Sector	BAU scenario, $MtCO_2e$	Emissions reduction in LTS4CN scenario, $MtCO_2e$	Emissions balance in LTS4CN scenario, $MtCO_2e$
Agriculture	34.9	-15.6	19.3
Energy	82.7	-54.3	28.2
FOLU	21.2	-71.4	(-50.2)
IPPU	10.7	-9.1	1.6
Waste	6.5	-5.3	1.2
Total	156.0	155.6	0,3

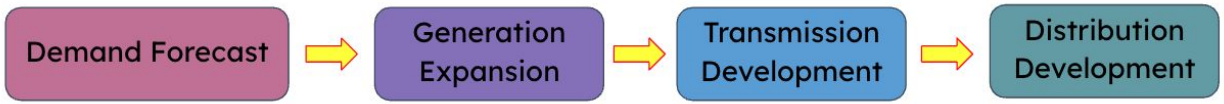
❖ Policy - PDP

(Energy)Lab

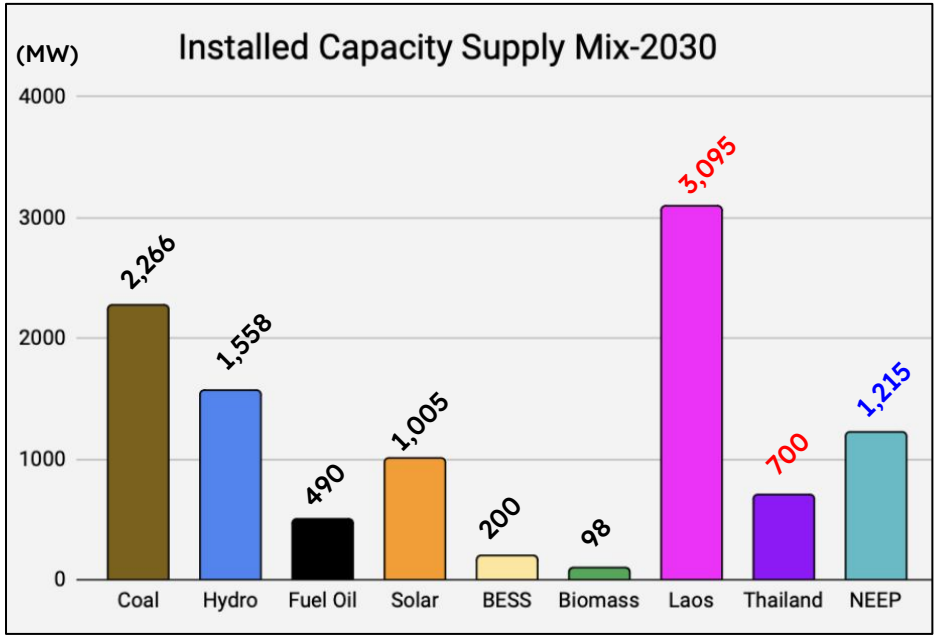
Power Development Plan (PDP) 2022-2040: A long-term plan for the development of the energy sector in Cambodia. A strategic framework designed to enhance electricity infrastructure, ensuring reliable, affordable, and sustainable energy.



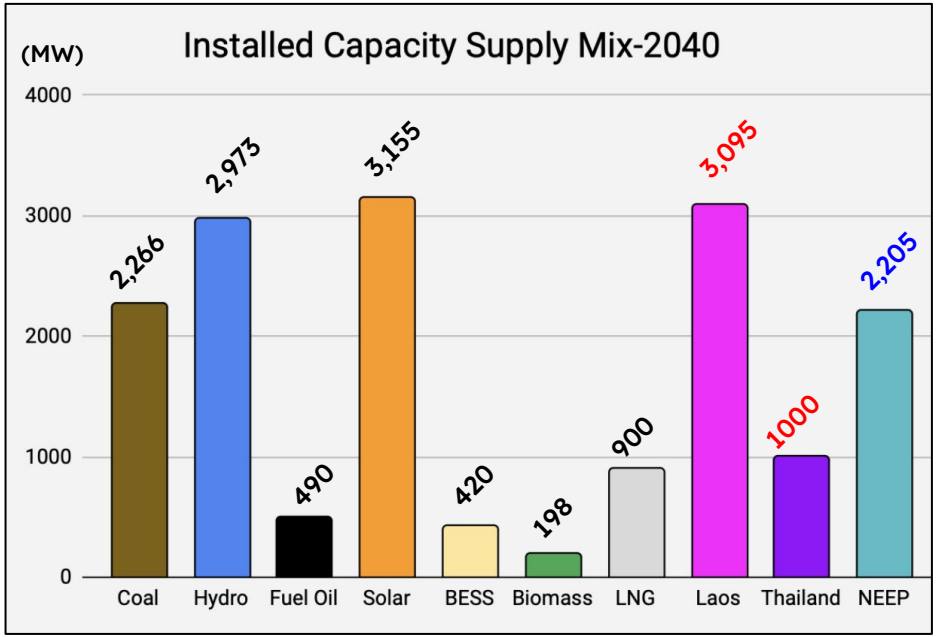
- ❖ Fulfill future energy demand
- ❖ Reduce dependency on energy imports
- ❖ Increase renewable energy



Installed capacity 2030



Installed capacity 2040

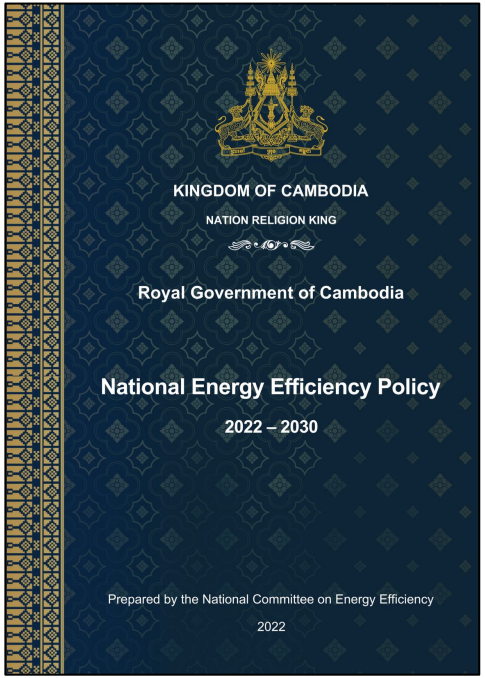


PDP Scenario 4

❖ Policy - NEEP

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Energy Efficiency = Using less energy to achieve the same level of output or performance, effectively minimizing energy waste and promoting responsible resource management.



19% total energy consumption reduction by 2030:

- ❑ 20% from Industrial Sector
- ❑ 34% from Residential Sector
- ❑ 25% from Commercial Building
- ❑ 29% from Public Service
- ❑ 5% from Transport Sector



❖ Policy - NEEP

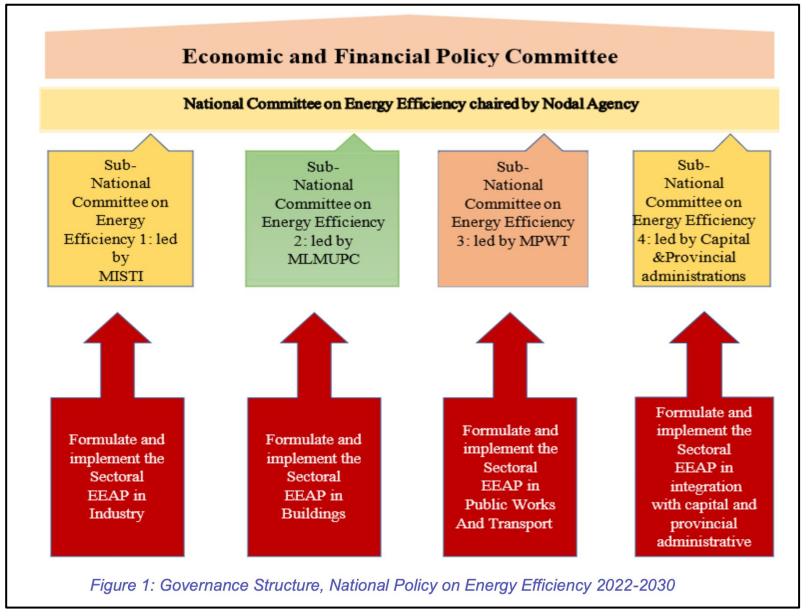
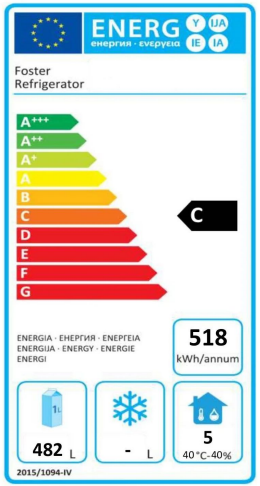


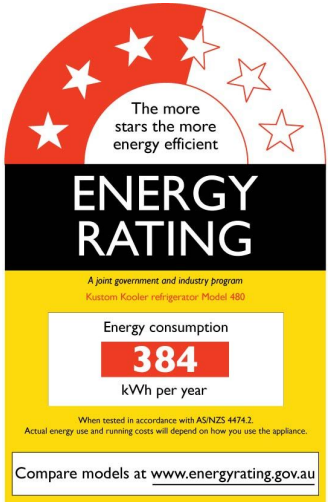
Figure 1: Governance Structure, National Policy on Energy Efficiency 2022-2030

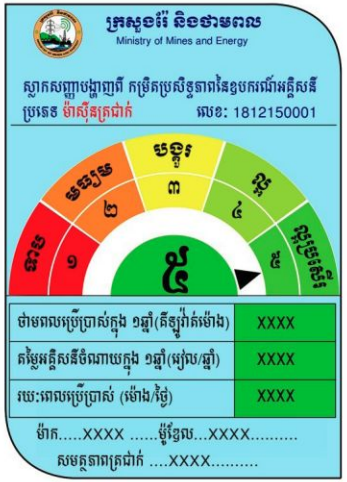
Intervention to achieve the target:

- **Industrial sector:** Energy Management Program, Energy-efficient technologies (machinery & processes)
- **Commercial Sector:** Energy efficiency standards for building, incentives for energy-efficient equipment, energy audits, etc.
- **Residential Sector:** Minimum Energy Performance Standards, Building Energy Code
- **Public services:** Energy efficiency for public buildings, change to LED for road lighting, etc.
- **Transport sector:** Transitioning to EVs, fuel efficiency, and public transportation

❖ Policy - NEEP


ENERG
Foster
Refrigerator
A+++
A++
A+
A
B
C
D
E
F
G
C
518
kWh/annum
482 L
- L
5
40°C-40%
2015/1094-IV


The more stars the more energy efficient
ENERGY RATING
A joint government and industry program
Kustom Kooler refrigerator Model 480
Energy consumption
384
kWh per year
When tested in accordance with AS/NZS 4474.2.
Actual energy use and running costs will depend on how you use the appliance.
Compare models at www.energyrating.gov.au


Ministry of Mines and Energy
លេខ: 1812150001
ចំណាយលប្រើប្រាស់ក្នុង ១ឆ្នាំ(គិតជាដុល្លារ/ម៉ាង)
XXXX
តម្លៃអគ្គិសនីចំណាយក្នុង ១ឆ្នាំ(រៀល/ឆ្នាំ)
XXXX
រយៈពេលប្រើប្រាស់ (ម៉ោង/ថ្ងៃ)
XXXX


energy
ENERGY STAR

❖ Policy - EV

(Energy)Lab

Transport sector - one of the largest direct contributors to GHG.

The National Policy on the Development of EV 2024-2030 serves as a strategic roadmap and driving force to attract investments, **promote EV adoption**, and develop the entire EV ecosystem in Cambodia.


Kingdom of Cambodia
Nation Religion King

Royal Government of Cambodia

National Policy on the Development of
Electric Vehicles 2024 - 2030

May, 2024

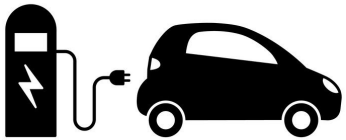


720,000 E2Ws



20,000 E3Ws

2030 GOAL

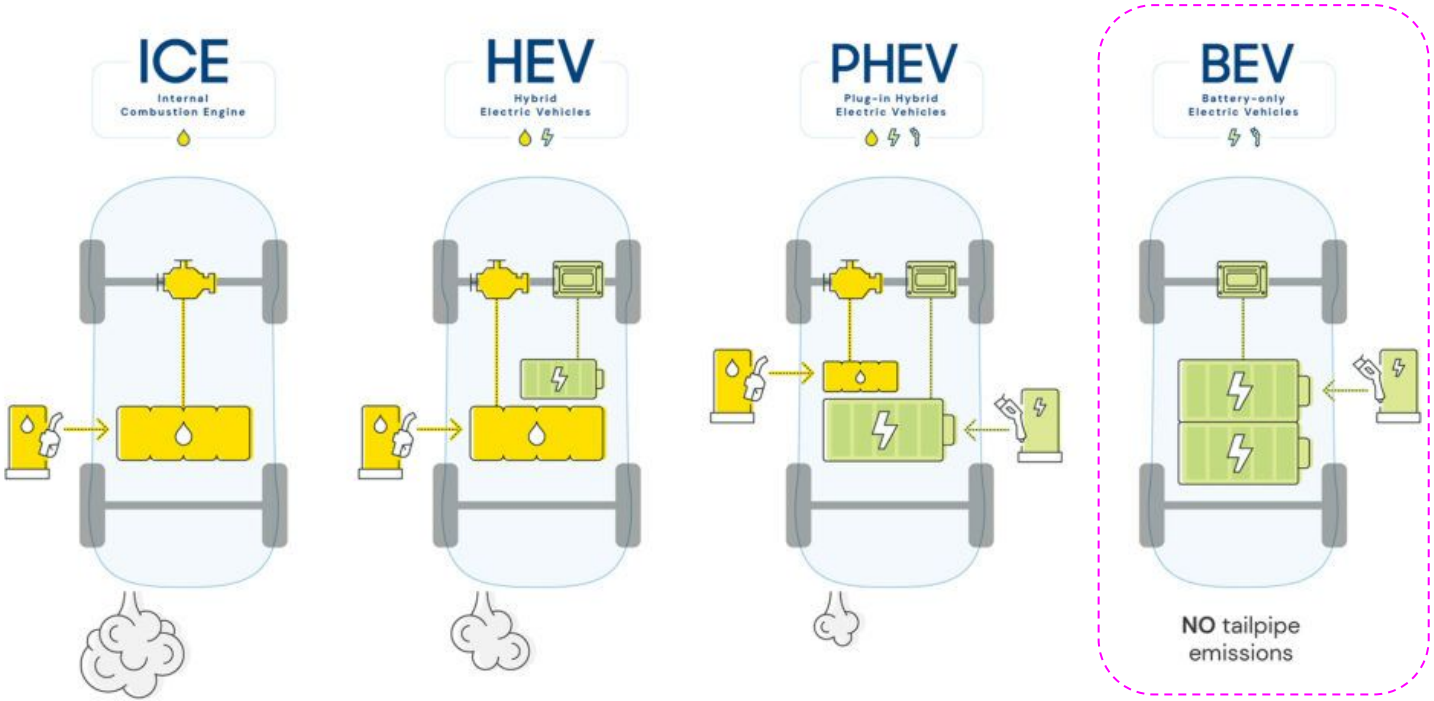


25,000 E4Ws



5,000 Heavy-duty vehicles

❖ Policy - EV


ICE
Internal Combustion Engine
HEV
Hybrid Electric Vehicles
PHEV
Plug-in Hybrid Electric Vehicles
BEV
Battery-only Electric Vehicles
NO tailpipe emissions

Key measures to achieve the target



Development of
Supporting infrastructure



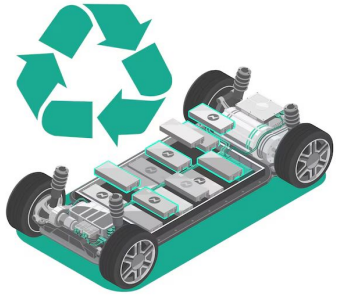
Establishment of
Supply chain



Establishment of
Incentive mechanism



Awareness raising



EoL waste management

Incentive Mechanism: Tax Exemption

Car Import Duties	
ICE Car	*BEV*
Custom Duties: 35%	Custom Duties: 35%
Special Duties: 50%	Special Duties: 10%
VAT: 10%	VAT: 10%
Total ~ **120%**	Total ~ **60%**

Electric Vehicle

An opportunity or threat?

Sub-Decree No. 18, February 2021
Special Tax Adjustment

Sub-Decree No. 52, March 2026
Import Tax Adjustment

*ICE & HEV Car***	*EV (BEV only)*	*EV (BEV)*	*EV (PHEV)*
Import Tax: 35%	Import Tax: 35%	Import Tax: **0%**	Import Tax: **7%**
Special Tax: 60% >> **50%**	Special Tax: 30% >> **10%**	Special Tax: 10%	Special Tax: 10%
VAT: 10%	VAT: 10%	VAT: 10%	VAT: 10%
Total ~ **120%** *of the vehicle price*	Total ~ **60%** *of the vehicle price*	Total ~ **21%** *of the vehicle price*	Total ~ **29.5%** *of the vehicle price*

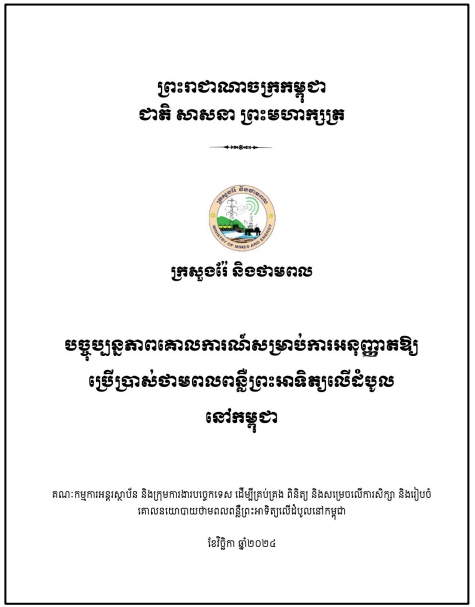
** Cylinder size/equivalent engine power: From 1,000 cc to 3,000 cc

Adjustment of customs import duty (Sub-Decree No. 52)

Items	Previous Rate	Rate Effective 1 April 2026
EV Battery Charger, Electric Rice Cooker, Solar Lamps/Lights	7%	0%
Solar PV System, Electric Cookstove, Electric Kettle, Commercial EV (BEV, HEV, PHEV), Motor for EV, Lithium Battery	15%	0%
Family PHEV	35%	7%
Family BEV and Induction Cooker	35%	0%

** EV: Electric Vehicle*
** BEV: Battery Electric Vehicle — Vehicles than run on battery only.*
** HEV: Hybrid Electric Vehicle — Vehicles than run on a combustion engine with a backup battery pack (usually small). Generally, the vehicles can only be refuled while the battery packed cannot be charged using chargers.*
** PHEV: Plug-in Hybrid Electric Vehicle — Vehicles than run on both combustion engine and battery pack (usually large). The vehicles can be refuled while the battery pack can be charged using recommended chargers.*
** Electric Cookstove (ចង្ក្រានកម្តៅអគ្គិសនី): Uses electricity to heat coils or radiant element, heating the surface first before the cookware that sits on it.*
** Induction Cooker (ចង្ក្រានអគ្គិសនី): Uses electromagnetism to heat cookware directly, offering fasted, safest, and most precise cooking.*

Principle for Permitting the Use of Rooftop Solar (RTS Regulation)



ព្រះរាជាណាចក្រកម្ពុជា
ជាតិ សាសនា ព្រះមហាក្សត្រ

ក្រសួងរ៉ែ និងថាមពល

បច្ចុប្បន្នភាពគោលការណ៍សម្រាប់ការអនុញ្ញាតឱ្យ
ប្រើប្រាស់ថាមពលពន្លឺព្រះអាទិត្យលើដំបូល
នៅកម្ពុជា

គណៈកម្មការអន្តរស្ថាប័ន និងក្រុមការងារបច្ចេកទេស ដើម្បីគ្រប់គ្រង ពិនិត្យ និងសម្រេចលើការសិក្សា និងរៀបចំ
គោលនយោបាយថាមពលពន្លឺព្រះអាទិត្យលើដំបូលនៅកម្ពុជា

ខែវិច្ឆិកា ឆ្នាំ២០២៤

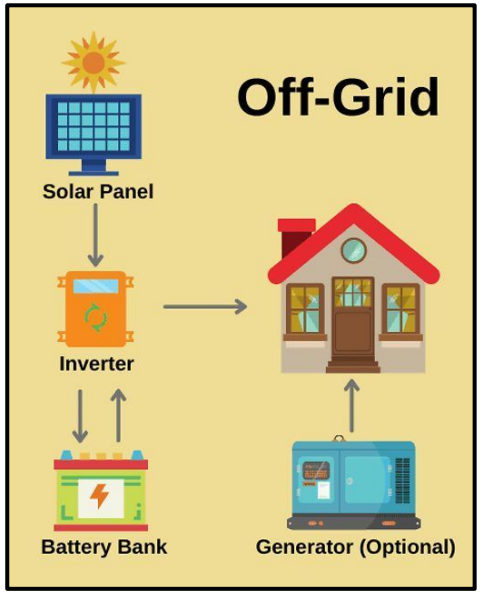
RTS refers to solar PV system installed behind the consumer's energy meter with the premise of the consumer for **self-consumption** purposes.

This could be installed either on the **roof** or on the **land** of the property.

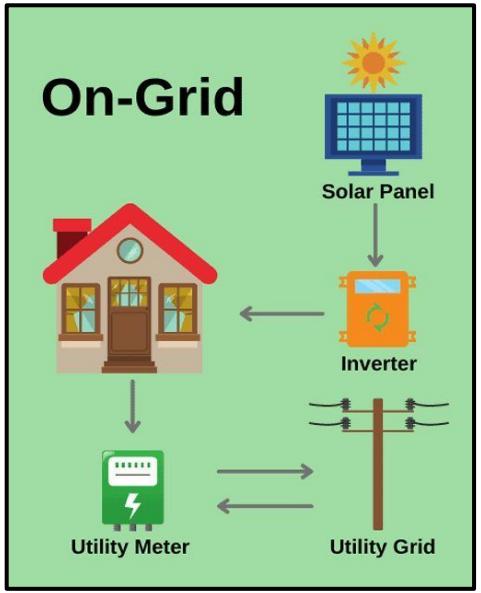




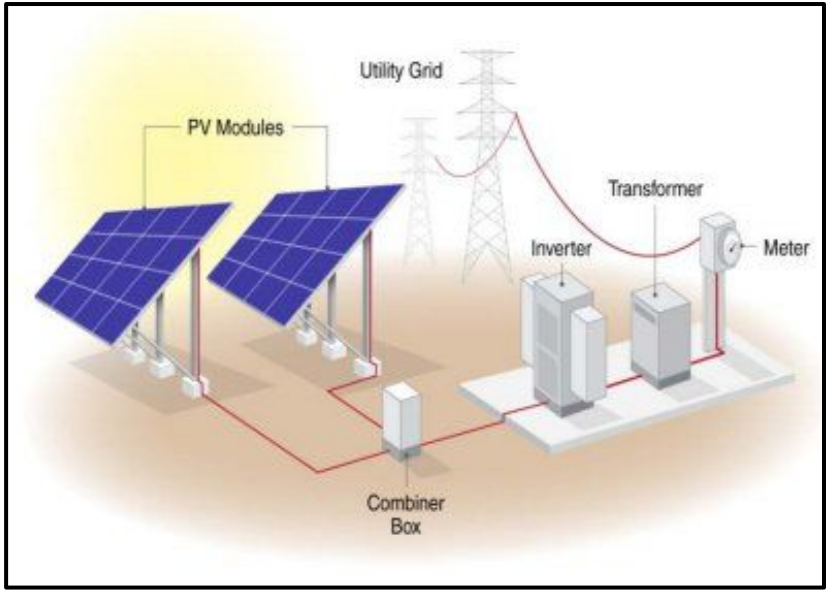
Off-grid PV



On-grid PV



Utility-scale PV



❖ Policy - Revised RTS

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Ministry of Mines and Energy (MME)

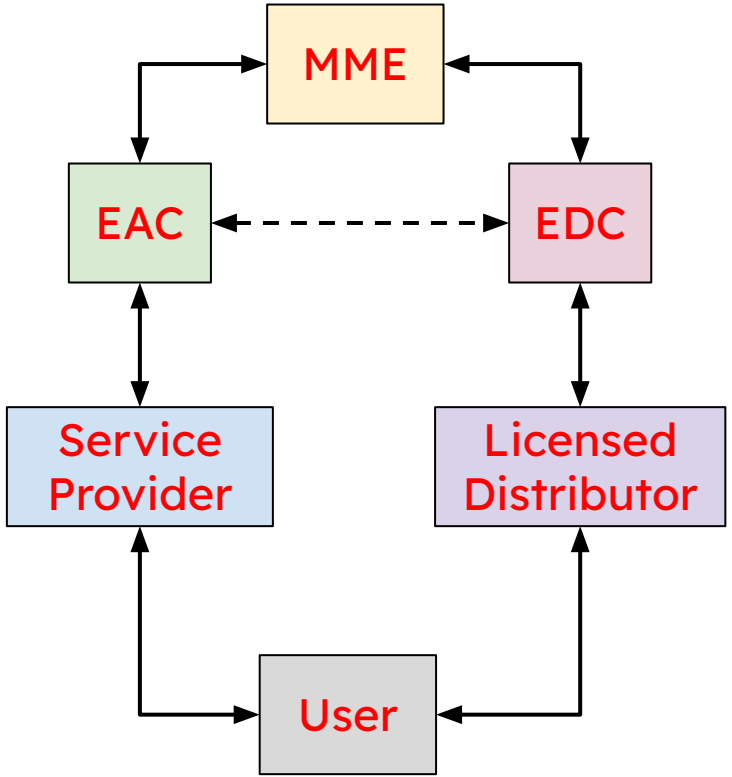
- Policy & Regulation
- Guidelines & Standards
- Quota setting
- Implementation monitoring

Electricity Authority of Cambodia (EAC)

- Licenses issuing
- Service provider managing
- Quota establishing
- Permit providing
- Tariff setting

Electricité du Cambodge (EDC)

- Support quota setting
- Support guidelines and standards
- Testing and commissioning of solar system
- Analyze impacts on the grid
- Billing



❖ Policy - Revised RTS

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Purposes:

- Issue permit for RTS installation, ensuring safety for the national grid
- Allocate quota scheme to fulfill environmental obligations, aligning with the PDP
- Ensure technical standard and safety of the use of RTS

Key Features of the RTS:

- *Quota scheme:* Solar installation allocated based on quota defined from PDP
- *Project implementation:* Specific permit and installation procedures
- *Excess power feed-in:* RTS for self-consumption only. Feed-in allowed but with no credit
- *Licensing:* Service providers must have a license issued by EAC
- *Registration:* All on-grid systems to be registered
- *Tariff:* Compensation tariff payable to EDC for energy generated and used from solar system.
- *Others:* No more capacity restriction, no PPA, no third-party ownership

Compensation Tariff

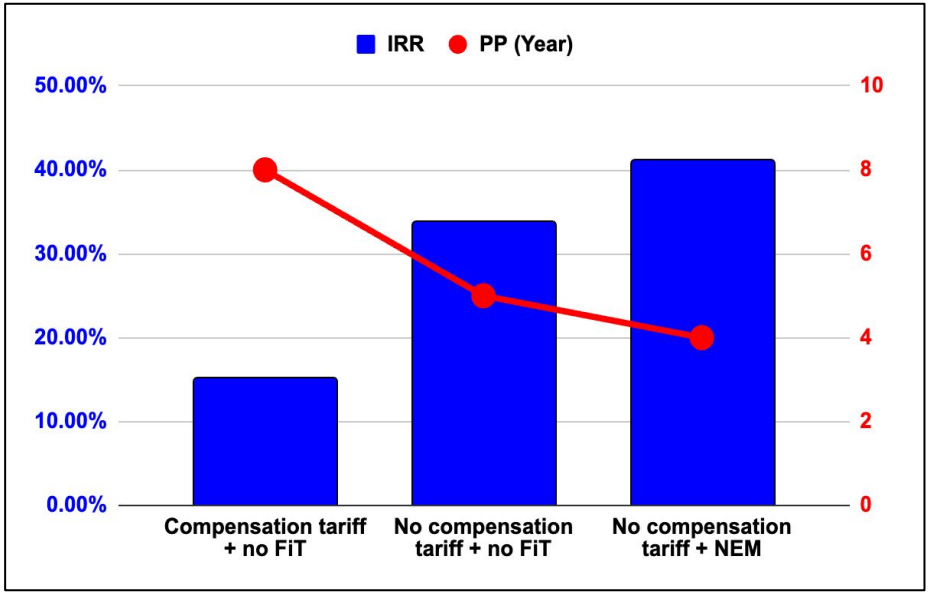
Tariff payable to EDC for energy generated and used by the solar owner, compensating the EDC for energy that would have otherwise been purchased.

	Installed capacity of rooftop solar PV	Compensation tariff
Small	Less than 10 kW	No compensation tariff
Medium	From 10 to 50 kW	$0.037/kWh
	From 50 to 100 kW	$0.047/kWh
Large	From 100 to 200 kW	$0.059/kWh
	From 200 to 500 kW	$0.062/kWh
	From 500 to 1000 kW	$0.065/kWh
	Over 1000 kW	$0.067/kWh

❖ Policy - Revised RTS

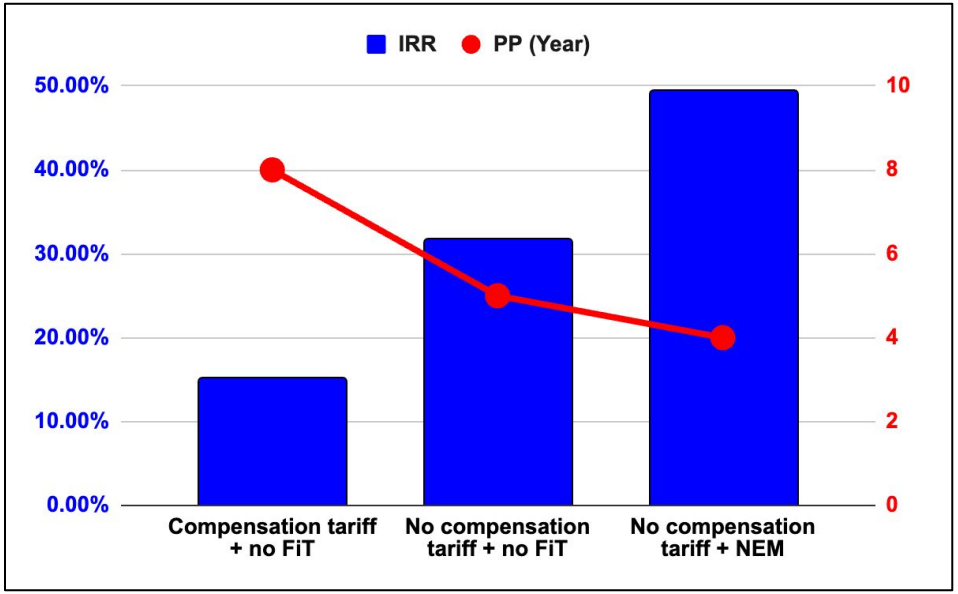
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Economic Analysis - Case Study



A garment factory

(1,000 kW)



An office building

(104 kW)

THANK YOU
FOR YOUR ATTENTION

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Materials